

Module 52 - async & await

1. What is async?

async is a keyword used before a function to make it return a Promise and allow the use of await inside the function.

2. What is await?

await is a keyword used inside an async function to wait for a Promise to complete before continuing with the next statement.

Example Code:

```
let newPromise = new Promise((fullFilled, failure) => {  
  let dataReceived = true;  
  if(dataReceived){  
    fullFilled("Data Fetched Successfully");  
  }  
  else{  
    // failure("Data Not Found");  
    throw new Error("Search Proper Data");  
  }  
});  
  
async function executePromise(){  
  try{  
    let message = await newPromise;  
    let newMessage = await newPromise;  
    console.log(message);  
    console.log(`Next Message: ${newMessage}`);  
  }  
  catch(error){  
    console.log(error.message);  
  }  
  finally{  
    console.log("End");  
  }  
}  
  
executePromise();  
console.log("Last");
```

3. What is Exception Handling?

Exception Handling is a mechanism used to detect, handle, and manage errors during program execution without stopping the entire program.

- **try** is used to write code that may cause an error.
- **catch** is used to handle the error that occurs inside the **try** block.
- **finally** is used to execute code after **try** and **catch**, whether an error occurs or not.
- **throw** is used to manually create and send an error to the **catch** block.

Example Code:

```
try {  
    let age = 15;  
    if (age < 18) {  
        throw new Error("You are not eligible.");  
    }  
    console.log("You are eligible.");  
}  
catch (error) {  
    console.log("Error: " + error.message);  
}  
finally {  
    console.log("Execution completed.");  
}
```